

Motivation 3 — MIO: exact best subset

Solve the discrete problem directly

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Draft outline — content to come.

The mixed-integer formulation

$$\min_{\beta_0, \beta, z} -\frac{1}{n} \ell(\beta_0, \beta) \quad \text{s.t.} \quad z_j \in \{0, 1\}, \quad \sum_j z_j \leq k, \quad |\beta_j| \leq M z_j.$$

What MIO gets right

- Exact solution of the original ℓ_0 problem.
- Branch-and-bound with modern solvers (Gurobi, CPLEX, `bestsubset`).
- No relaxation bias.

Where MIO breaks down

- **Computational cost** explodes beyond $p \gtrsim$ a few hundred.
- **Overfits** because there is no shrinkage — every selected coefficient is fully fit.
- Requires a careful Big- M choice.
- Multinomial / GLM extensions are even slower.

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